



Re-Engineering Climate Science Communication: The Future of IPCC Reporting

Abstract

Since 1990, the Intergovernmental Panel on Climate Change (IPCC) has produced major assessments of climate science about every six years. In 2013, as the fifth assessment was released, climate scientist Kevin Trenberth argued that the IPCC should declare success and change its approach. This lesson explores how IPCC reports are made, what they achieved, and why a faster, continuous stream of climate information may now be needed.

Introduction

The IPCC's role is to provide scientific advice that is relevant to policy but does not tell governments what to do. Its reports shaped world events: in 1995 it concluded that the balance of evidence suggested a discernible human influence on climate, which fed into the Kyoto Protocol. In 2007, it stated that warming was unequivocal and very likely due to human activities, and later that year the IPCC shared the Nobel Peace Prize. But producing these reports is a huge, unpaid effort by volunteer scientists, and society's needs have changed. Now that climate change is here, Trenberth asked whether waiting six or seven years between reports still makes sense.

How Do Scientists Study This?

Each IPCC report reviews all the published literature from the previous five to seven years. Authors assess the state of knowledge, reconcile disagreements, resolve discrepancies, and document uncertainties. The drafts go through two major reviews: first by experts, then by governments. Climate skeptics can and do take part, some even as authors. Every comment receives a written response, and review editors oversee each chapter. Because United Nations rules require seeking unanimous consensus, the final wording is negotiated for accuracy, balance, and clarity. Scientists decide what can be said; governments help decide how it is said. Trenberth analyzed this process to weigh its strengths against its costs.

How an IPCC assessment is built



What Did They Discover?

Trenberth concluded that the consensus process is a strength, but it also makes IPCC reports conservative, and the effort is huge, cumbersome, and burdensome for volunteer scientists. Because the climate of the past is no longer a good guide to the future, society needs a continuous stream of information on how and why climate is changing. He proposed that the separate working groups on physical science and on impacts and adaptation should be combined. Routine updates could come from annual reports such as the Bulletin of the American Meteorological Society's State of the Climate, which ran to 238 pages for 2012, and from climate services like the World Meteorological Organization's Global Framework on Climate Services.

IPCC milestones

1990

first IPCC assessment report

1995

discernible human influence on climate

2007

warming unequivocal; Nobel Peace Prize

Trenberth suggested targeted, rigorously reviewed reports, like those of the U.S. National Research Council, as a model for answering governments' specific questions.

Takeaways

IPCC succeeded

Its reports built global agreement that humans are changing the climate.

Consensus has costs

Seeking unanimous agreement makes reports reliable but slow and conservative.

Keep information flowing

Climate services can provide continuous, practical updates for planning.

Key Terms

IPCC

The Intergovernmental Panel on Climate Change, a United Nations body that assesses climate research for policymakers.

Policy-relevant, not policy-prescriptive

Providing information useful for decisions without telling decision-makers which choice to make.

Consensus

General agreement among all participants. IPCC wording must be accepted by all member governments.

Working Group

A section of the IPCC focused on one area, such as physical science or impacts and adaptation.

Climate services

Ongoing provision of climate information and tools to help people plan and make decisions.

Match-Up: draw a line from each term to its clue

A IPCC

B Policy-relevant, not policy-prescriptive

C Consensus

D Working Group

E Climate services

1 Ongoing provision of climate information and tools to help people plan and...

2 General agreement among all participants

3 A section of the IPCC focused on one area

4 Providing information useful for decisions without telling decision-makers which choice to make

5 The Intergovernmental Panel on Climate Change, a United Nations body that assesses...

Answer key: A-5 · B-4 · C-2 · D-3 · E-1

Activity

Redesign the Report

Students weigh the strengths and weaknesses of the IPCC process and propose a better way to share climate science.

Materials: T-chart worksheet; Sticky notes; Lesson text

1. Create a T-chart listing strengths and weaknesses of the IPCC process described in the lesson.
2. Identify two groups who need climate information, such as city planners or farmers, and how often they need updates.
3. Design a new reporting system that keeps at least two IPCC strengths while fixing one weakness.
4. Explain how your system would stay trustworthy if it moved faster.
5. Share your design and vote on which idea best balances speed and reliability.

References

Trenberth, K. (2013). Time to change how the IPCC reports? The Conversation.
<https://doi.org/10.64628/AA.rg4fq6hmf>